Database Systems	
Lab #:	**05**
Lab Title:	Custom calculations in queries and aggregate function.
Submitted by:	

Names	**Registration #**
Sumaira Safeer	FA22-BCE-019

Rubrics name & number		**Marks** In-Lab	**Marks** Post-Lab
Engineering Knowledge	***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.*		
Problem Analysis	***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.*		
Design	***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.*		
Modern Tools Usage	***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.*		
Individual and Teamwork	***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.*		
	R13: Management of Team Work: *Ability to appreciate, understand and work with multidisciplinary team members.*		

Rubrics #	**R2**	**R6**	**R8**	**R9**	**R12**	**R13**
In Lab						
Post- Lab						

Lab No: 05

Title: Custom calculations in queries and aggregate function.

Objective:

Using a query makes it easier to view, add, delete, or change data in your database. The objective of query is:

- Find specific quickly data by filtering on specific criteria (conditions).
- Calculate or summarize data.
- Automate data management tasks, such as reviewing the most current data on a recurring basis.
- Write queries with aggregate functions like COUNT, MAX, MIN, and SUM Advance Queries:
- Custom calculation in queries.

Inlab Tasks:

- **bktblAuthor:**

Design View:

bktblAuthor

Field Name	Data Type
ID	AutoNumber
Name	Short Text
email	Short Text
Contact No	Short Text
address	Short Text

Datasheet View:

ID	Name	email	Contact No	address
1	G Thomsan	gewqjk21@gmail.com	304281918	Canada
2	RW Robot	qwguu30@gmail.com	301346932	London
3	Robert frost	gwqui49@gmail.com	308233278	Japan
4	Pugpne Goldstein	wekge58@gmaol.com	305231788	Abudhabi
5	Bil Gates	ftyu67@gmaol.com	345677889	Pakistan
(New)				

- **bktblPUBLISHER:**

Design View:

bktblAuthor | **bktblPUBLISHER**

Field Name	Data Type
ID	AutoNumber
BOOK NAME	Short Text
DATE OF PUBLISH	Short Text

Field Properties

General | Lookup

Field Size	Long Integer
New Values	Increment
Format	

Datasheet View:

bktblAuthor | **bktblPUBLISHER**

ID	BOOK NAME	DATE OF PUBLISH
1	Vile Bodies	23-02-2003
2	Wuthering heights	14-02-2008
3	A Little women	13-12-2017
4	The greatest	29-05-2004
5	A Scanner Darkly	14-02-2023
6	Jane Eyre	10-09-2024
(New)		

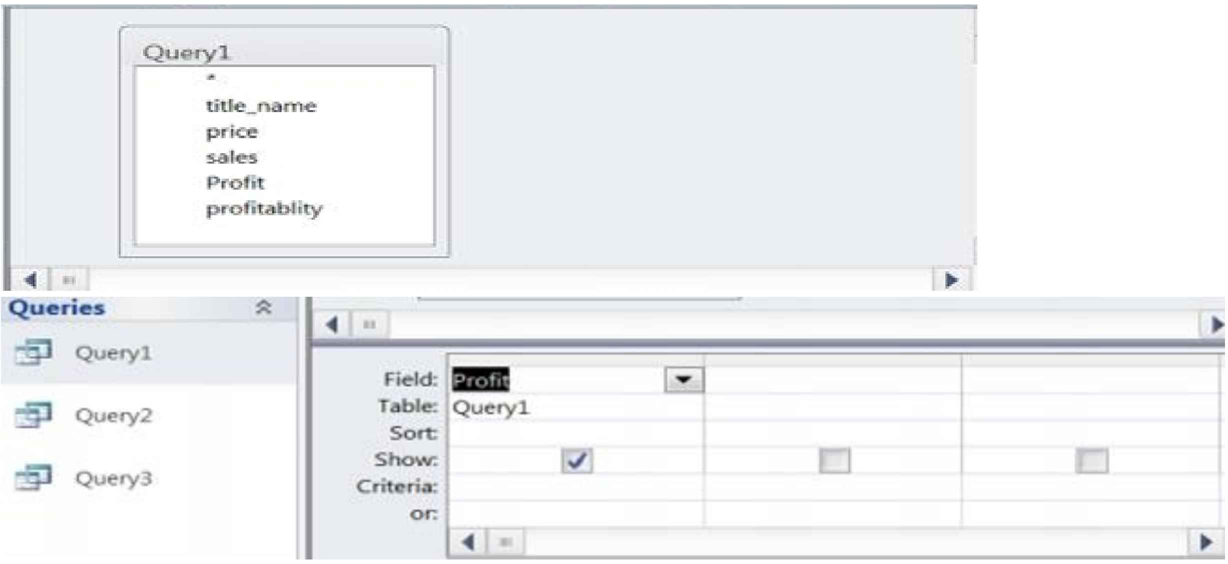
Table:

TitleID	PublisherID	TitleName	Type	Pages	Price	Sales	PubDate	Contract	Advance	AuthorID
1	1	The Great Nove	Fiction	300	$125,000.00	500	1/1/2020	yes	10000	1
2	2	Mystry Book	Thriller	250	$12,000.00	3000	1/6/2019	No	0	2
3	3	Science fiction	Sci-Fic	400	$30.00	7000	1/9/2020	yes	15000	1
(New)										

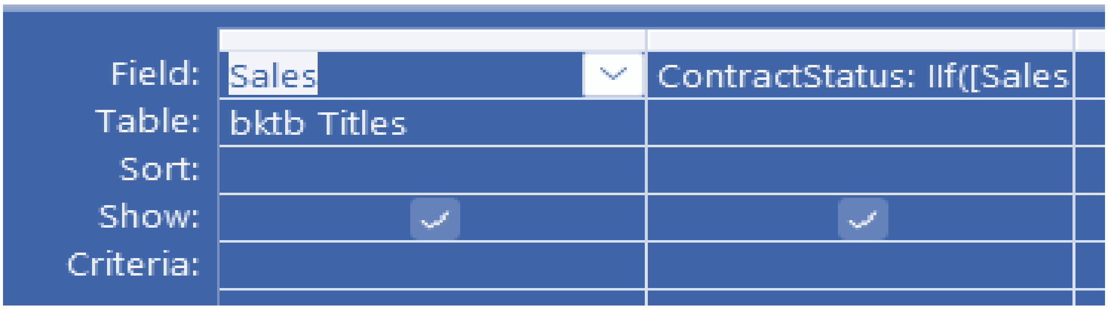
Task 1: (Simple if else statement)

We would like our new field to contain the value “has a contract” if the value of the sales field in bktblTitles is greater than 500, and “no contract” if that value is less than and equal to 500. Write the expression in expression builder.

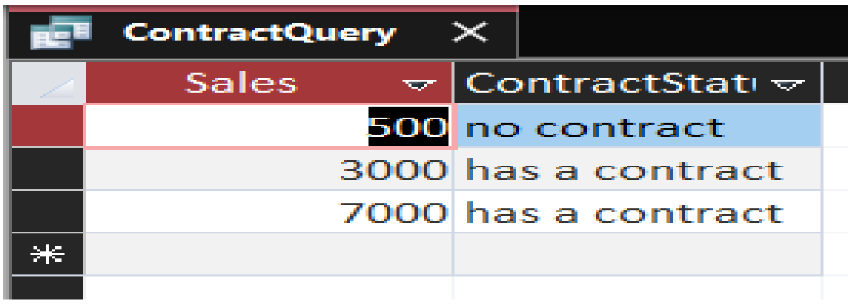
Creating a Query:



Query:



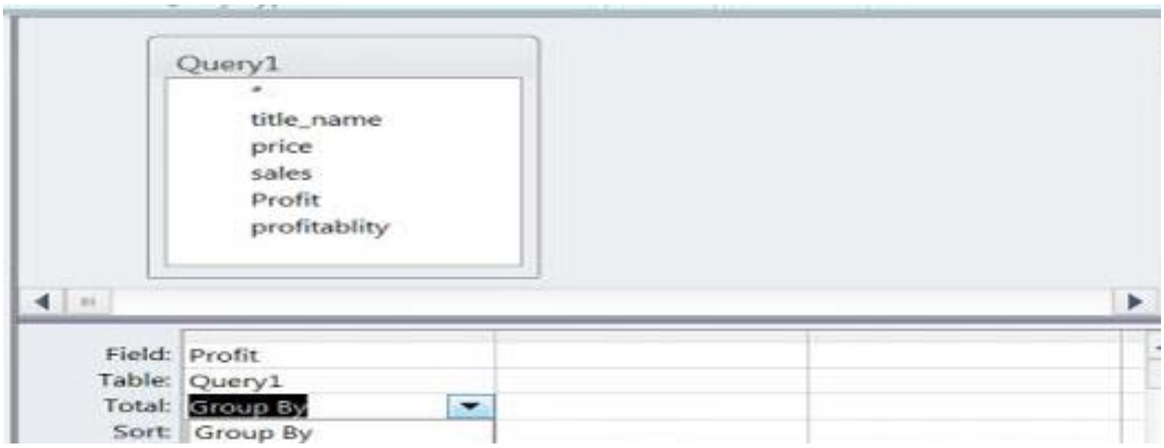
Query Result:



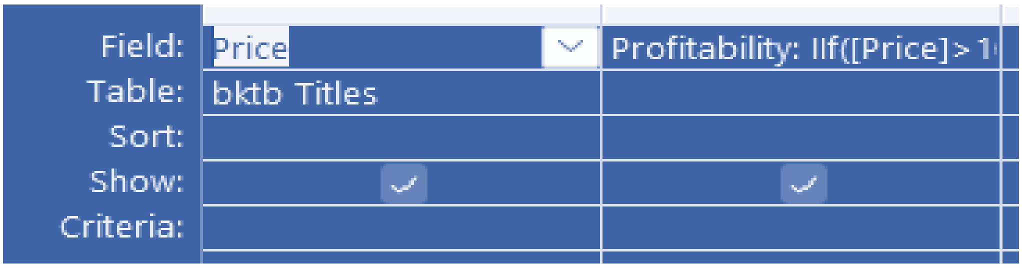
Task 2: (Nested if else statement)

Create a new field on query 1 called profitability. This field should display "very profitable" if the price was greater than 10,000; "profitable" if the price was less than or equal to 10,000 and greater than 5,000; “OK” if the price was less than or equal to 5,000 and greater than 1,000; and “not enough” otherwise.

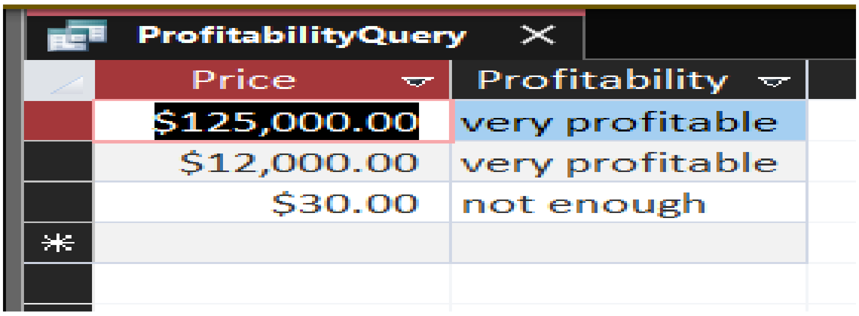
Query Design View:



Query:



Query Result:



Task 3:

Create a new query and count the total number of books that have been published.

Books Record:

Query1

ID	Name	email	BOOK NAME	Contact No	Expr1005	DATE OF PUBLISH
1	G Thomsan	gewqjk21@gmail.com	Vile Bodies	304281918	23-02-2003	23-02-2003
2	RW Robot	qwguu30@gmail.com	Wuthering heights	301346932	14-02-2008	14-02-2008
3	Robert frost	gwqui49@gmail.com	A Little women	308233278	13-12-2017	13-12-2017
4	Pugpne Goldstein	wekge58@gmaol.com	The greatest	305231788	29-05-2004	29-05-2004
5	Bil Gates	ftyu67@gmaol.com	A Scanner Darkly	345677889	14-02-2023	14-02-2023

<u>Query Result:</u>

TotalBooksQuery

TotalBooks
3

Task 4:

Create a Query so that it shows the number of books each publisher has published.

Query:

Field:	PublisherName	BooksPublished: TitleNa
Table:	bktblPublishers	bktblTitles
Total:	Group By	Count
Sort:		

<u>Query Result:</u>

BooksPublished

PublisherNan	BooksPublish
Hachette Book	1
Harper Collins	1
Penguin Randon	1

Task 5:

Create query so that it shows the count of books of type “programming” OR “networking” for each publisher.

Field:	PublisherName	ProgrammingAndNetwo	[bktblTitles].[BookType]
Table:	bktblPublishers		
Total:	Group By	Expression	Where
Sort:			
Show:	✓	✓	☐
Criteria:			"Programming" Or "Net
or:			

Query Result:

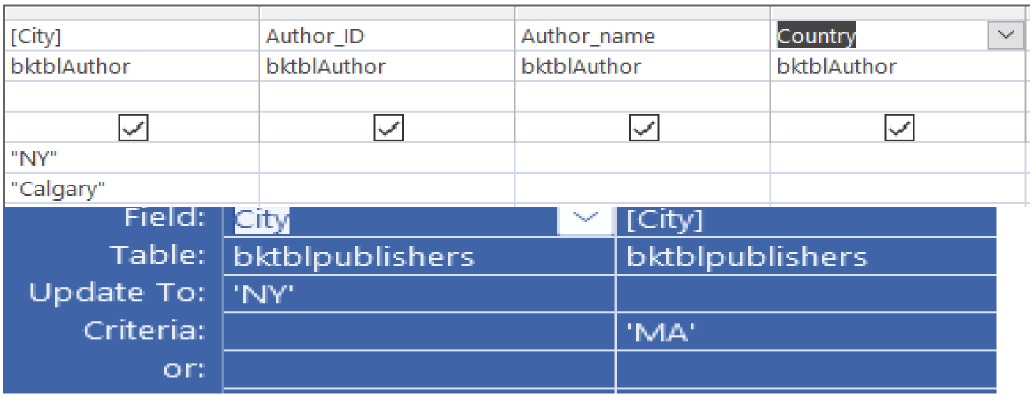
ProgrammingOrNetworking

PublisherNan	Programming
Hachette Book	1
Harper Collins	1
Penguin Randon	1

Post Lab Tasks:

Task 1:

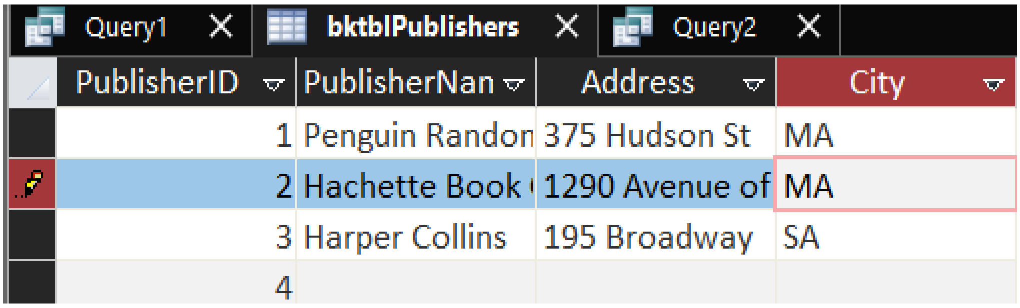
Update the city field to 'NY' for all records in the bktblpublishers table where the City is “MA".



[City]	Author_ID	Author_name	Country
bktblAuthor	bktblAuthor	bktblAuthor	bktblAuthor
☑	☑	☑	☑
"NY"			
"Calgary"			

Field:	City	[City]
Table:	bktblpublishers	bktblpublishers
Update To:	'NY'	
Criteria:		'MA'
or:		

Query Result:

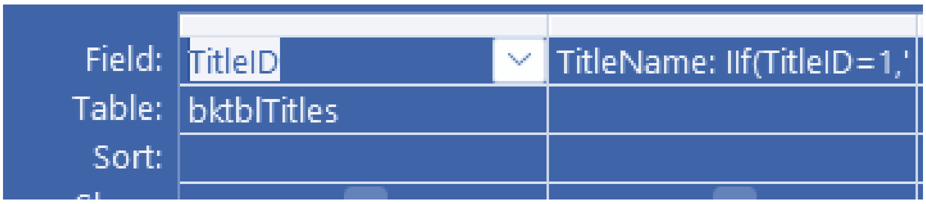


PublisherID	PublisherNan	Address	City
1	Penguin Randon	375 Hudson St	MA
2	Hachette Book (	1290 Avenue of	MA
3	Harper Collins	195 Broadway	SA
4			

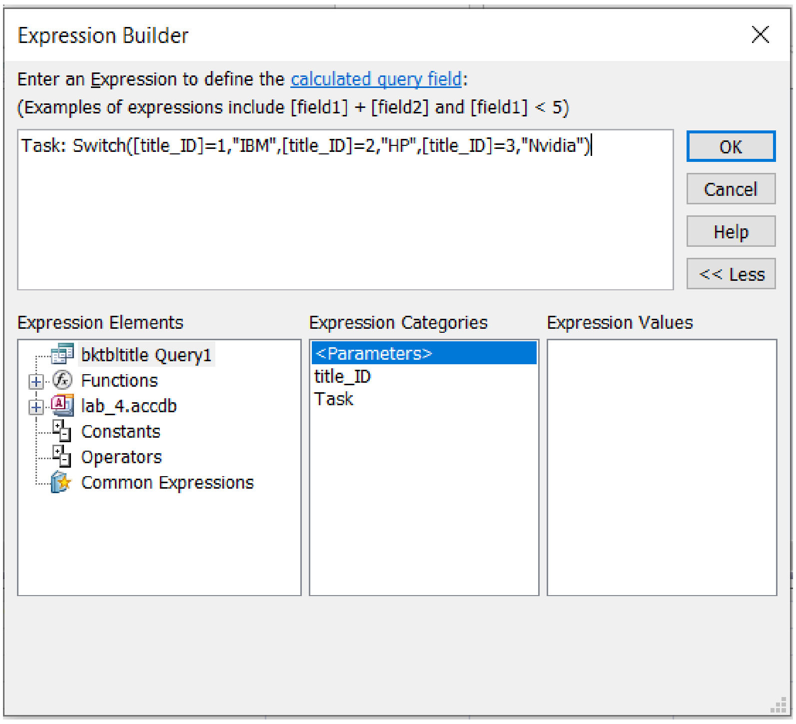
Task 2: (Switch Function)

In the bktbltitle:

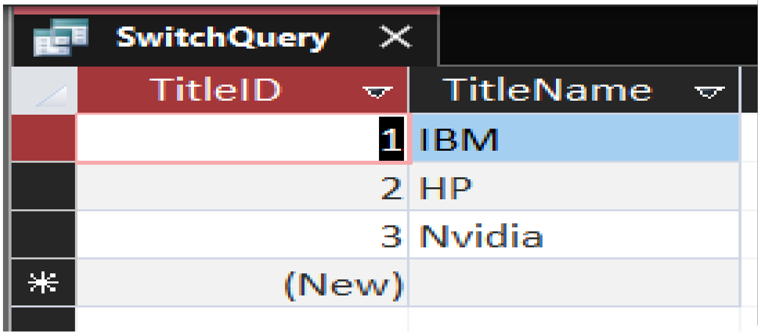
- If the TitleID field is 1, then the Switch function will return "IBM".
- If the TitleID field is 2, then the Switch function will return "HP".
- If the TitleID field is 3, then the Switch function will return "Nvidia".

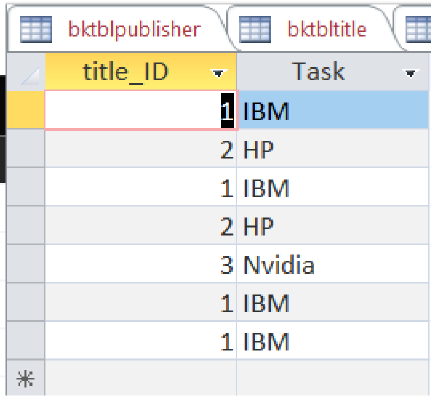


Field:	TitleID	TitleName: IIf(TitleID=1,'
Table:	bktblTitles	
Sort:		



Query Result:

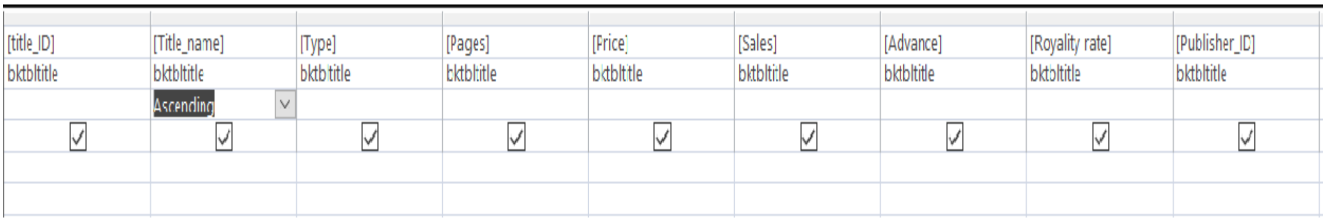




Task 3:

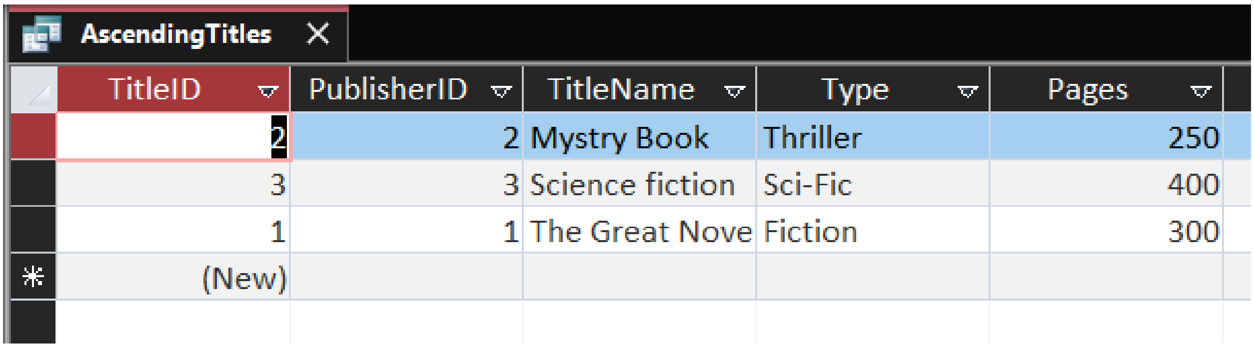
Create and run a query that sorts the results alphabetically, A-Z (Ascending) by title_name and then change it to show them in reverse alphabetical order (Descending).

1. Ascending Order



Field:	TitleName
Table:	bktblTitles
Sort:	Ascending
Show:	
Criteria:	

<u>Query Result:</u>



AscendingTitles

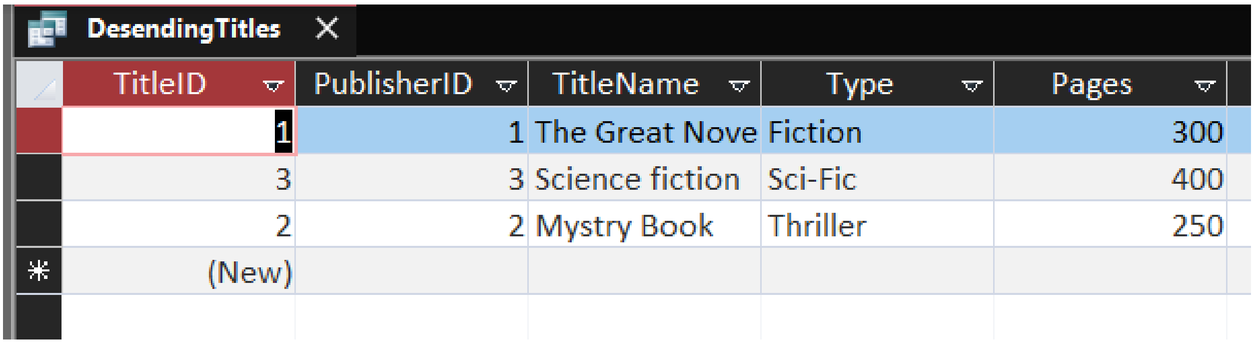
TitleID	PublisherID	TitleName	Type	Pages
2	2	Mystry Book	Thriller	250
3	3	Science fiction	Sci-Fic	400
1	1	The Great Nove	Fiction	300
(New)				

2. Descending Order

[title_ID]	[Title_name]	[Type]	[Pages]	[Price]	[Sales]	[Advance]	[Royality rate]	[Publisher_ID]
bktbltitle	bktbltitle	bktbltitle	bktbltitle	bktbltitle	bktbltitle	bktbltitle	bktbltitle	bktbltitle
	Descending							
☑	☑	☑	☑	☑	☑	☑	☑	☑

Field:	TitleName
Table:	bktblTitles
Sort:	Descending
Show:	

<u>Query Result:</u>



DesendingTitles

TitleID	PublisherID	TitleName	Type	Pages
1	1	The Great Nove	Fiction	300
3	3	Science fiction	Sci-Fic	400
2	2	Mystry Book	Thriller	250
(New)				